MILITARY SPECIFICATION

PLASTIC SHEETS, LAMINATED, THERMOSETTING, ELECTRICAL INSULATING SHEET, GLASS-MAT

1. SCOPE

1.1 Scope. This specification covers laminated plastic sheets for mechanical and electrical applications.

1.2 Classification. Plastic sheets covered by this specification shall be of the grade and classes specified in the specification sheet.

1.2.1 Grade. The grade is identified by a symbol GPO-N-. The last digit will be the number of the applicable specification sheet.

1.2.2 Class. The plastic sheet shall be furnished in the following temperature classes as specified in the applicable specification sheet. The class of the plastic sheet indicates the temperature index in degrees Centigrade (°C.), (see 6.1).

Class
130
155
180
200

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

FEDERAL
FED-STD-406 - Plastics: Method of Testing.

MILITARY
MIL-STD-104 - Limits for Electrical Insulation Color.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 149 - Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies.
D 150 - A-C Loss Characteristics and Dielectric Constant (Permittivity of Solid Electrical Insulating Materials.
D 229 - Rigid Sheet and Plate Materials Used for Electrical Insulation.
D 374 - Thickness of Solid Electrical Insulation.
D 495 - High-Voltage, Low-Current Arc Resistance of Solid Electrical Insulating Materials.
D 618 - Conditioning Plastics and Electrical Insulating Materials for Testing.
D 790 - Flexural Properties of Plastics.
D 877 - Dielectric Breakdown Voltage of Insulating Liquids using Disk Electrodes.
D 2303 - Liquid-Contaminant, Inclined-Plane Tracking and Erosion of Insulating Materials.
D 2304-64T - Thermal Evaluation of Rigid Electrical Insulating Materials.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103.)

Institute of Electrical and Electronics Engineers (IEEE)
Standard No. 98 - Thermal Evaluation of Electrical Insulating Materials, Guide for the Preparation of Test Procedures.

(Application for copies should be addressed to the Institute of Electrical and Electronics Engineers, 345 East 47 Street, New York, New York 10017.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheets, the latter govern.

3.2 Qualification. Plastic sheets furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.3 Construction. The material shall be made from a mat of random-laid glass fibers which is saturated with a resin combined with suitable fillers, cured by heat and pressure and processed to conform to this specification.

3.4 Machinability. The material shall be such that it can be drilled, tapped, sawed, and machined in all directions in accordance with the manufacturer's recommeded technique without cracking, splitting, or otherwise impairing the material for general use.

3.5 Color. Plastic sheets used in electrical equipment rated at less than 2000 volts shall be natural in color unless otherwise specified (see 6.2). Natural color is the color produced by natural undyed base, or filler and resin used. Plastic sheets used in high voltage electrical equipment rated at 2000 volts or higher shall be in accordance with class 1 of MIL-STD-104.

3.6 Surface defects. The material shall be free from blisters, wrinkles, or cracks, and reasonably free from other small defects such as scratches, dents, heat marks, and so forth.

3.7 Uniformity. All sheets of corresponding thickness shall be uniform in texture, finish, and specified properties.

3.8 Surface finish. The surface finish of sheets shall be semi-gloss as produced by the laminating operation, unless otherwise specified by the procuring activity (see 6.2).

3.9 Marking. Each full size sheet furnished for stock shall be legibly marked with the manufacturer's name or trade mark, grade, and class. The method of marking shall be in accordance with MIL-STD-130.

3.10 Property values. The material shall conform to the property values specified in the individual specification sheets (see 4.3 and 4.4). The values obtained for each set of specimens taken from the same sheet shall be averaged before comparison with the applicable table.

3.11 Dimensions and Tolerances.

3.11.1 Thicknesses. Sheets shall be furnished in the nominal thicknesses shown in table I, as specified (see 6.2). The average thickness of the sheet shall be within the variations specified in the table and at no point shall the thickness as measured vary from the nominal thickness by a value greater than the individual tolerance specified in table I.

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Table I - Thickness 1/

Nominal Thickness		Permissible Variations
Inches	Decimal Equivalent In inches	In Inches (±)
1/16	0.0625	0.008
3/32	.094	.009
1/8	.125	.010
5/32	.156	.011
3/16	.1875	.012
7/32	.219	.014
1/4	.250	.015
5/16	.3125	.018
3/8	.375	.020
7/16	.438	.022
1/2	.500	.024
5/8	.625	.027
3/4	.750	.029
7/8	.875	.031
1	1.000	.033
1-1/8	1.125	.035
1-1/4	1.250	.037
1-3/8	1.375	.039
1-1/2	1.500	.041
1-5/8	1.625	.043
1-3/4	1.750	.045
1-7/8	1.875	.047
2	2.000	.049

1/ On sheets of nominal thickness not listed in table I, the permissible variation shall be the same as for the next greater thickness.

3.11.2 Length and width. Unless otherwise specified in the contract or order, the manufacturer's standard sizes between 24 and 48 inches in width and 36 and 96 inches in length will be acceptable. The length and width of trimmed sheets may vary 1 inch over or under the manufacturer's standard size. Nonuniform standard sheet dimensions caused by cutting specimens for tests required by this specification shall not be cause for rejection, unless particular dimensions are specified. When particular sheet dimensions are specified (see 6.2), the permissible variations from the specified length or width shall be as specified in table II.

Table II - Permissible variations in length and width. (Cut sizes)

Nominal Thickness	Permissible Variations in length or width, in inches (±)		
INCHES	6 INCHES AND UNDER	OVER 6 to 18 INCHES, INCL.	Over 18 to 24 INCHES, INCL.
Up to 1/4, incl.	0.010	0.015	1/32
5/16 to 1/2, incl.	.012	.017	1/32
5/8 to 1, incl.	.015	.020	1/32
1-1/8 to 1-1/2, incl.	.018	.030	1/16
1-5/8 to 2, incl.	.022	.040	1/16

3.12 Warp or twist. (Applicable only to individual manufacturer's standard size sheets). The warp or twist of material, as delivered shall not exceed that specified in table III (see 4.7.4).

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Table III - Warp or twist.

Nominal thickness	Permissible variation, on basis of 36 inch dimension
Inches	Maximum percent
1/16 to 1/8	2.50
5/32 to 1/4	1.50
5/16 to 3/4	0.50
7/8 and over	0.25

3.13 Workmanship. The laminated sheets shall be manufactured and processed in a careful and workmanlike manner. The resin shall be completely bonded to the glass-mat and shall show no exposed reinforcement.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchast order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and testing of all laminated plastic sheets covered by this specification shall be classified as follows:

(a) Qualification inspection.
(b) Quality conformance inspection.

4.2.1 The tests applicable to each classification shall be as shown on table IV:

Table IV - Categories of inspection.

Examination or test	Test paragraph	Qualification inspection	Quality conformance inspection
Appearance and workmanship	4.4.4.1.1		X
Dimensional defects	4.4.4.1.2		X
Preparation for Delivery	4.4.4.1.3		X
Thickness	4.7.3	X	
Warp and twist	4.7.4		X
Dielectric breakdown parallel to laminations	4.7.5	X	X
Dielectric constant	4.7.6	X	
60 hertz			
1 megahertz			
Dissipation factor	4.7.6	X	
60 hertz			
1 megahertz			
Arc resistance	4.7.7	X	
Wet tracking	4.7.8	X	X
Impact strength, tested edgewise	4.7.9	X	
Flexural strength, tested flatwise (crosswise)	4.7.10	X	X
Bond strength	4.7.11	X	X
Water absorption	4.7.12	X	
Flame resistance	4.7.13	X	X
Ignition time			
Burning time			
Thermal endurance	4.7.14	X	
Silicone content	4.7.15	X	
Machinability	4.7.16	X	

4.3 Qualification.[1/] Qualification inspection shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification inspection shall consist of the examinations and tests shown in table IV as specified in the applicable specification sheet.

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4.4 Quality conformance inspection. Quality conformance inspection shall be performed on each lot offered for delivery as specified in the applicable specification sheet.

4.4.1 Lot. For purposes of sampling, a lot shall consist of all material of the same grade, class, thickness, length, width, and color offered for delivery at one time.

4.4.2 Sampling for quality conformance inspection. Sampling for quality conformance inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.4.3 Rejection. Any sample failing to conform to any test shall be rejected and if the rejection number exceeds the acceptance number, the lot represented by the sample shall be rejected.

4.4.4 Examination. Examination of the laminate shall be made in accordance with the classification of defects, inspection levels, and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105 shall be expressed in units of sheets, for the examinations of 4.4.4.1.1, 4.4.4.1.2, and in units of shipping containers for the examination of 4.4.4.1.3.

4.4.4.1 Inspection levels and acceptable quality levels for examination. The inspection levels for determining the sample size and the AQL's expressed as defects per 100 units as specified in MIL-STD-105 shall be as follows:

Examination paragraph	Inspection level	AQL
4.4.4.1.1	II	1.5
4.4.4.1.2	S-4	2.5
4.4.4.1.3	S-2	4.0

4.4.4.1.1 Examination of the laminate for defects in appearance and workmanship. Each sheet in the sample selected from the lot in accordance with 4.4.4.1 shall be examined as follows:

Examine	Defect
Appearance	Not uniform texture or finish. Blisters, wrinkles, cracks, splits, or holes. Excessive number of scratches, dents, or heat marks. Presence of dirt, foreign material, or embedded particles.
Workmanship	Not semi-gloss finish. Not natural in color or red. Reinforcement exposed, not uniformly spaced. Incomplete bond.
Markings	Not marked as specified.

4.4.4.1.2 Examination of laminate for dimensional defects. Each sheet in the sample selected from the lot in accordance with w.w.w.1 shall be examined as follows:

Examine	Defect
Length and width (Standard sizes)	Varies by more than 1 inch over or under manufacturer's standard size.
Length and width (cut sizes)	Varies by more than tolerance indicated in table II.
Thickness	Varies by more than the tolerances indicated in table I. (The average thickness shall be within tolerances specified with no one measurement exceeding tolerance indicated in table I for applicable thickness).

4.4.4.1.3 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking comply with the requirements of Section 5 of this specification. Each shipping container in the sample selected from the lot in accordance with 4.4.4.1 shall be examined. The shipping containers shall be fully packed and

1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3 and 6.4).

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selected just prior to the closing operation. The same number of containers fully prepared for delivery shall be examined for closure defects.

Examine	Defect
Packaging	Not level specified. Sheets not unit wrapped or interleaved; protective wrapping. Not as specified.
Packing	Not level specified. Construction and material not as specified. Container not as specified, closures not made by specified or required methods or materials. Pads or liners not provided.
Count	Less than specified or indicated quantity of sheets.
Weight	Gross weight exceeds specified requirements.
Markings	Interior or exterior markings (as applicable) omitted, illegible, incorrect, incomplete or not as specified herein or in the contract or order.

4.5 Quality conformance testing. Quality conformance testing shall be conducted in accordance with the individual specification sheets and applicable paragraphs of this specification for the characteristics as indicated herein on each lot submitted. The sample unit shall be one sheet having at least two square feet of area with 18 inches minimum dimension. The sample size shall be S-1 as specified in MIL-STD-105. The acceptable quality level shall be 4.0 expressed in defects per 100 units. The lot shall be expressed in units of sheets.

4.6 Conditioning of test specimens.

4.6.1 Nomenclature. The designations indicating conditioning of test specimens shall be in accordance with ASTM D 618.

4.6.2 Time tolerances for conditioning. The time tolerance for conditioning shall be as specified in table V.

Table V - Time tolerance for conditioning.

Conditioning		Cooling		Comments
Condition	Time Tolerance	Time	Time Tolerance	
48/50	-0 hours +2 hours	16 hours or more	-0 hours	Cool in a dry desiccator. Start test within 1/2 hour after removing specimen from desiccator.
48/50/water 24/23/water 336/50/water	-0 hours +1/2 hour	1 hour	-0 hours	Cool by immersion in a sufficient quantity of distilled water to reduce the temperature to 23°C. within 1 hour. Remove individually as needed, and wipe surface water off with a cloth. Start test within 1 minute after removing specimen from water.
1/105	-0 minutes +6 minutes	2 hours or more.	0 hours	See water absorption test. (4.7.12)
1/105 24/23/water	-0 hours 2 hours	---	---	See water absorption test. (4.7.12)

4.6.3 Temperature and relative humidity tolerances. The conditioning temperature and relative humidity tolerances shall be in accordance with ASTM D 618.

4.7 Test procedures.

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4.7.1 Definitions. In conducting tests, the term "lengthwise" shall be interpreted to mean that sheet direction known to be the stronger in flexure; "crosswise" shall then be the sheet direction known to be the weaker in flexure.

4.7.2 Test conditions. Unless otherwise specified, test on the conditioned specimenns shall be made at a room temperature of 23°C. ± 3°C.

4.7.3 Thickness. Thickness shall be measured in accordance with ASTM D 374 at 10 points distributed over the entire sheet. On test specimens, the dead weight dial micrometer, method C, shall be used. On sheets with 1 dimension over 24 inches method B, shall be used. A micrometer fitted to a yoke of sufficient size and rigidity to permit accurate measurements in the center of the sheet shall be used (see 3.11).

4.7.4 Warp and twist. The warp and twist test shall be made on a full size sheet, a portion of which may be used for destructive tests. The warp and twist of the sheets shall be measured in accordance with ASTM D 229 (see 3.12).

4.7.5 Dielectric break-down parallel to laminations (step-by-step).

4.7.5.1 Specimens. The test specimens shall be 2 by 3 inches by the thickness of the sheet. American standard taper pins, such as Morse, Brown and Sharpe, or Pratt and Whitney, or their equivalent having a taper of 1/4 inch per foot, shall be used. For test specimens having a thickness up to 1/2 inch, inclusive, No. 3 American standard taper pin [2/] 3 inches long and having a diameter at the large end of 7/32 inch shall be used. Drill two 3/16 inch diameter holes, centrally located, 1 inch apart center to center, and perpendicular to the faces of the specimen. Ream holes to a sufficient depth to allow the pins to extend approximately 1-1/2 inch from the small end of the hole. The electrodes shall be inserted after the conditioning of the test specimen. Spheres having a 1/2 inch diameter, when placed on the extremities of the tapered pins, will decrease the tendency to flash over.

4.7.5.2 Electrical apparatus. Step up transformer, circuit breaker, voltage control, and voltmeter shall be in accordance with ASTM D 149. Power supply frequency shall be not greater than 100 Hertz, the transformer shall have a rating of not less than 5 KVA.

4.7.5.3 Procedure. All tests shall be made under oil, at commercial power frequencies of 60 Hertz. One specimen only shall be tested by the short-time method for the purpose of providing a basis for the initial voltage applied in the step-by-step test. In the short-time test, starting at zero, the testing voltage shall be increased as uniformly as possible at a rate of 500 volts per second. On the remaining specimens, the step-by-step method shall be used. The initial voltage applied in the step-by-step test shall be 50 percent of the short time breakdown voltage. The initial voltage in the step-by-step tests determined as specified in 4.7.5.1 shall be applied for 1 minute and the voltage shall then be increased in increments as follows, holding the voltage at each step for 1 minute:

Breakdown voltage by short-time method (kilovolts)	Increments of increase for step-by-step method (kilovolts)
25 or less	1.0
Over 25 to 50	2.0
Over 50 to 100	5.0
Over 100	10.0

4.7.5.4 Oil medium breakdown. If the specimen should appear to fail or fall below a specified minimum failure voltage and there is doubt as to whether the breakdown occurred in the oil or in the specimen, the oil itself shall be tested as specified in ASTM D 877. If the breakdown voltage of the oil, tested at room temperature is found to be less than 22 KV, the oil shall be changed or reconditioned so that the breakdown voltage is not less than 22 KV.

4.7.5.5 Report. The breakdown voltage shall be reported in kilovolts, and the time required to break each specimen shall be reported in seconds.

4.7.6 Dielectric constant and dissipation factor.

4.7.6.1 Test equipment. Bridge, generator and detector shall be as described in ASTM D 150 for low frequency measurements.

[2/] For information on tapered pins, see Kents Mechanical Engineers Handbook, 12 Edition, Design and Production page 15-14. (This handbook may be obtained from Wiley and Sons, Inc., 440 4th Ave., New York, New York 10016.)

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4.7.6.2 Specimens. The specimens for dielectric constant and dissipation factor tests shall be 4 inch squares cut from 1/8 inch in thickness.

4.7.6.3 Procedure. Specimens shall be removed from the water bath individually as needed and all surfaces wiped dry with a cloth. Metal foil electrodes 0.001 inch thick shall be immediately applied to both sides of the specimen using a thin layer of petrolatum as an adhesive. The top electrode shall be 2-5/8 inches in diameter and the bottom electrode shall be 3 inches in diameter carefully rolled on to exclude air bubbles. As an alternate electrode system, conductive silver paint may be used when placed on the specimen prior to conditioning. Measurements shall be completed within 1/2 hour after removal from water at 60 Hertz and 1 Megahertz.

4.7.6.4 Calculations. Calculations shall be made in accordance with the procedures outlined in ASTM D 150.

4.7.7 Arc resistance. The arc resistance test shall be performed in accordance with ASTM D 495 using Tungsten rod electrodes.

4.7.8 Wet tracking. Track resistance test shall be performed in accordance with ASTM D 2303 time to track method using liquid contaminant on specimens at least 1/4 inch in thickness, samples less than 1/4 inch in thickness shall be plied up to at least 1/4 inch thick, in the "as received" condition.

4.7.9 Impact strength (tested edgewise). The impact test shall be performed in accordance with ASTM D 229.

4.7.10 Flexural strength (tested edgewise). The flexural strength test shall be performed at room temperature in accordance with ASTM D 790.

4.7.11 Bond strength. Bond strength shall be performed in accordance with ASTM D 229 on a standard specimen 1 inch square by 0.500 $\pm$ 0.005 inch thick.

4.7.12 Water absorption.

4.7.12.1 Specimens. The specimens shall be 3 inches by 1 inch by thickness.

4.7.12.2 Procedure. The specimens shall be subjected to a temperature of 105°C. for 1 hour, cooled in a desiccator for at least 2 hours, and weighed (W_1). The specimens shall then be immersed in distilled water for 24 hours at 23°C. They shall be removed individually as needed, all surfaces wiped off with a dry cloth, and then reweighed immediately (W_2). The test shall begin not more than 1 minute after removal of the specimens from the water and shall be completed as rapidly as consistent with accuracy.

4.7.12.3 Calculations. The percentage of water absorption shall be calculated as follows:

$$\text{Percentage of water absorption in 24 hours} = \frac{W_2 - W_1}{W_1} \times 100$$

4.7.13 Flame resistance. Flame resistance shall be determined in accordance with method II of ASTM D 229.

4.7.14 Thermal endurance and temperature classification. Thermal endurance and temperature classification shall be conducted in accordance with sections 4.2.2, 6, 7, 8, and 10 of ASTM D 2304.

4.7.14.1 End points. The thermal life (end points) shall be conducted in accordance with section 5 ASTM D 2304 and shall be not less than 50 percent of the "as received" flexural strength value.

4.7.14.2 Calculations. The thermal life (end points) of each exposed temperature shall be plotted on graph paper having as ordinate a logarithmic time scale and an abscissa arranged according to the reciprocal of the absolute temperature. If applicable, draw a straight line by the use of regression analysis to represent the life of the material as a function of temperature. The temperature classification index shall be determined by the intersection of the Life-Temperature line with the 25,000 hour ordinate.

4.7.14.3 Alternate data. Thermal aging data obtained prior to the date of this specification and obtained using procedures other than ASTM D 2304 will be considered for acceptance by the Naval Ship Engineering Center, provided such data was obtained using the IEEE Standard No. 98 and the testing laboratory is acceptable to the Naval Ship Engineering Center, Materials Development and Applications Office.

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4.7.15 Silicone content for classes 130 and 155. Volatile siloxanes shall not be in excess of 25 ppm as determined by the gas chromatographic method.

4.7.15.1 Chromatographic method for the detection of siloxane vapors.

4.7.15.1.1 Sample size. The specimen size shall be 10 inches by 1 inch by the thickness.

4.7.15.1.2 Conditioning procedure. The sample shall be placed in a 250 ml. Erlenmeyer flask, fitted with a ground glass stopper containing a 2 mm. diameter capillary. This assembly shall then be maintained in an air oven for 168 hiurs at 185°C.

4.7.15.1.3 Sampling and detection of vapors. A 10 ml. sample of vapor in the Erlenmeyer flask shall be obtained by inserting a syringe needle through the capillary in the glass stopper. A convenient sampling device consists of a 10 ml Luer-lock syringe fitted with a 4 inch 20 or 22 gauge hypodermic needle. Next the sample shall be injected into a gas chromatograph and its chromatogram recorded.

4.7.15.1.4 Chromatographic column.

Description	Packed column	Capillary column
Stationary phase	Silicone oil-DC 550 20 percent	Silicone oil-DC 550
Support	C-22 Firebrick 42/60 mesh	Non-applicable
Length	6 feet	150 feet, min.
Diameter	1/8 inch	Standard
Oven temperature	70°C.	70°C.
Detector	Hydrogen flame ionization	Hydrogen flame ionization

4.7.15.1.5 Calibration. Calibrate chromatographic system using cyclic hexamethyltrisiloxane (D_3 trimer) in 1, 2-dichlorobenzene or other suitable solvent.

4.7.15.1.6 Qualitative detection. If any of the peaks on the chromatograms from the insulating materials show a retention time identical to that of the standard cyclic hexamethyltrisiloxane (D_3 trimer) degradative siloxane vapors are considered to be present; unless it can be shown by modified chromatographic techniques that this peak is not due to D_3 trimer. If no D_3 trimer. If no D_3 trimer is detected, a 10 ml. sample of vapor enriched with a small amount of D_3 trimer vapor shall be introduced into the column under identical conditions. Elution of the D_3 trimer at the appropriate retention time shall be considered proof of proper column functioning.

4.7.16 Machinability. A specimen of 1/2 inch thickness shall be drilled, tapped, sawed, and machined, in all directions, in accordance with the manufacturer's recommended technique, to determine conformance with 3.4.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements.)

5.1 Domestic shipment and early equipment installation.

5.1.1 Packaging. Packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include the grade, class, Federal stock number or manufacturer's part number, contract or order number, contractor's name, and destination.

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5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of packaging, packing, and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made (see 6.2).

5.2.1.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.2.1.1.1 Level A. Sheets shall be wrapped individually, or interleaved to protect them from abrasion, using kraft paper conforming to UU-P-268.

5.2.1.1.2 Level C. Packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to a supplier's commercial practice when such meets the requirements of this level.

5.2.1.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1.2.1 Level A. Sheets, packaged as specified (see 6.2) shall be packed in snug-fitting containers conforming to any one of the following specifications at the option of the supplier.

Specification	Type or class
PPP-B-591	Overseas type
PPP-B-621	Class 2
PPP-B-601	Overseas type

Boxes shall be provided with fiberboard pads and liners to protect edges and surfaces of sheets. The gross weight of boxes shall not exceed 200 pounds, unless the weight of a single sheet exceeds 200 pounds, in which case, sheets shall be packed one to a box.

5.2.1.2.2 Level B. Sheets, packaged as specified (see 6.2) shall be packed in containers conforming to class domestic of PPP-B-636. Shipping containers shall be closed in accordance with the box specification or appendix thereto. Fiberboard boxes shall not exceed the weight limitations of PPP-B-636.

5.2.1.2.3 Level C. Sheets, packaged as specified (see 6.2) shall be packed in containers in a manner which will insure acceptance by common carrier at the lowest rates and will afford protection against physical or mechanical damage during direct shipment from the supply source to the first receiving activity for immediate use. This level, in general, shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.2.1.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.)

6. Notes

6.1 Intended use. The plastic sheets covered by this specification are intended for use in electrical equipment where a rigid material must have good dielectric properties.

6.1.1 Temperature index. The temperature index is the highest temperature at which the material will provide satisfactory life as determined by the test method described herein. The temperature index is a guide and does not imply a thermal classification or a limitation on its use in electrical equipment.

6.1.2 Temperature classification. The temperature classification for electrical equipment should be determined in terms of thermal endurance of the insulation system rather than the individual materials.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Grade and class required.
(c) Color if other than as specified (see 3.5).
(d) Finish if other than as specified (see 3.8).
(e) Sheet size and thickness required (see 3.11).

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(f) Whether sheets, smaller in dimension than specified in the procurement document due to cutting specimens for lot acceptance tests are not acceptable (see 3.11.2).
(g) Packaging, packing, or marking requirements other than those required by 5.1 (see 5.2).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL-24364 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Center Building, Prince George's Center, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.4).

6.4 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.5 Fabricated parts or equipment. Requirements applicable to fabricated parts or equipment incorporating glass-mat resin laminated sheets should be specified in separate specifications or as part of contracts or orders.

Preparing activity:
Navy - SH
(Project 9330-N444)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0101-807-2000

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